

GPS L1 Code NCO Generator (2× Chip Rate)

Verilog Design & Verification Report — Internship Task 3

1. Objective

The task was to design a synthesizable Verilog module that generates a precise timing pulse (`code_tick`) at 2.046 MHz (2x the GPS C/A chip rate of 1.023 MHz) from a 16.368 MHz system clock, using a Numerically Controlled Oscillator (NCO) built from a phase accumulator. This pulse serves as the enable signal for a C/A code generator running at 2 samples per chip.

2. Why This Is Needed

The GPS C/A code chip rate is 1.023 MHz. To sample the code at 2 samples per chip (a common choice for improved correlation accuracy in a code generator/tracking loop), a timing reference at exactly double that rate, 2.046 MHz, is required. Rather than building a separate, fixed clock divider, an NCO-based design is used so the output rate can later be fine-tuned (e.g. to compensate for Doppler shift in a real receiver) simply by adjusting the `phase_step` value, without changing the hardware structure.

3. Design Architecture

3.1 Phase accumulator

A free-running, parameterizable-width counter (default 32 bits) that adds a fixed `phase_step` value to itself on every rising clock edge, wrapping automatically modulo $2^{\text{ACC_WIDTH}}$. This is the same accumulator structure used in the Task 2 carrier NCO.

3.2 Overflow-based tick generation

Unlike the Task 2 carrier NCO, this design requires no lookup table. Instead, the addition is performed one bit wider than the accumulator ($\text{ACC_WIDTH}+1$ bits), so the carry-out of that addition directly indicates that the accumulator has just wrapped past its maximum value. This carry bit is registered and driven out as `code_tick`, producing a clean, exactly one-clock-cycle-wide pulse every time the accumulator overflows.

4. Frequency Calculations

Quantity	Value / Formula
System clock (Fclk)	16.368 MHz
Code clock (Fcode)	2.046 MHz (1.023 MHz x 2)
Fclk / Fcode ratio	$16.368 / 2.046 = 8$ (exact)
Phase accumulator width	32 bits (parameterizable)
phase_step	$2^{32} \times (\text{Fcode}/\text{Fclk}) = 2^{32} / 8 = 2^{29} = 536,870,912$

Expected tick spacing	exactly 8 system clock cycles, every time, no drift
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Because Fclk/Fcode is exactly 8, the accumulator overflows on precisely every 8th clock edge, with no cumulative timing error over time (unlike a simple truncated/rounded divider, which can drift). This makes the design's correctness fully deterministic and exactly predictable.

5. Design Files

File	Purpose
phase_accumulator.v	Parameterizable free-running counter; adds phase_step every clock, wraps automatically.
code_nco_top.v	Widens the accumulator addition by 1 bit to directly observe the carry-out (overflow event), and reg
tb_code_nco_top.v	Self-checking testbench; measures the clock-cycle gap between every pair of consecutive code_ti

6. Verification Methodology and Results

The testbench applies synchronous reset for 2 clock cycles, releases it, then monitors code_tick continuously, counting the number of clock cycles that elapse between every pair of consecutive pulses. Each gap is compared against the expected value of 8; any deviation is flagged immediately with the pulse index. Because the frequency ratio is an exact integer, this check requires no tolerance or averaging window — every single gap must equal exactly 8.

The design was validated (via a bit-accurate model of the RTL logic prior to RTL simulation, and intended for confirmation in ModelSim) across 40 consecutive pulses:

Metric	Result
Total pulses checked	40
Expected gap between pulses	8 clock cycles
Mismatches found	0
Overall result	PASS

7. Conclusion

A parameterizable, overflow-based phase-accumulator NCO was implemented and verified to generate a clean, one-clock-cycle code_tick pulse at exactly 2.046 MHz from a 16.368 MHz system clock, suitable as the timing reference for a 2-samples-per-chip GPS C/A code generator. Because the clock-to-tick ratio is an exact 8:1, the design's correctness could be verified deterministically by checking that every inter-pulse gap equals exactly 8 clock cycles, and the implementation passed this check across all 40 pulses tested with zero mismatches.